

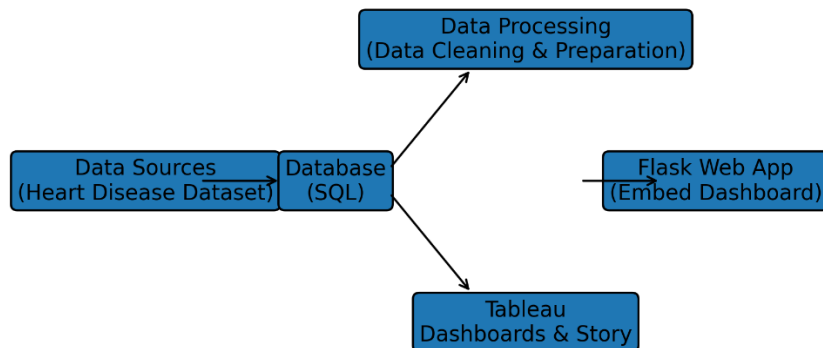
Project Design Phase-II Technology Stack (Architecture & Stack)

Date	6 th March 2026
Team ID	
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based dashboard for interactive health analysis.	HTML5, CSS3, JavaScript
2	Application Logic-1	Backend server handling routing and embedding logic.	Python (Flask Framework)
3	Application Logic-2	Business Intelligence logic for risk calculation and binning.	Tableau Desktop
4	Database	Structured storage of patient health indicators and lifestyle data.	MySQL
5	File Storage	Local storage for dataset extracts and Tableau workbooks.	Local Filesystem / .twb files
6	External Interface	Integration for embedding interactive visualizations into the web UI.	Tableau JavaScript API
7	Infrastructure	Local development and deployment environment.	Local Server Configuration

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Lightweight Python framework for web development and SQL connectors.	Flask, SQLAlchemy
2	Security Implementations	Secure routing and data sanitization to prevent unauthorized access to health data.	Flask-WTF, Werkzeug
3	Scalable Architecture	Decoupled 3-tier architecture allows independent scaling of the DB and the BI layer.	3-tier Architecture
4	Availability	Local hosting with optimized Tableau extracts ensures consistent access.	Tableau Hyper Engine
5	Performance	Use of Tableau's in-memory data engine for sub-second filtering of 300k+ rows.	Tableau Data Extracts

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>